

Got it  — no Streamlit, only local (clean + minimal)

Here is your correct DAY 14 (API only, no UI framework) 

DAY 14 — FASTAPI (LOCAL API ONLY)

GOAL

Your system runs as API

(No Streamlit, no frontend)

STEP 1 — INSTALL

pip install fastapi uvicorn

STEP 2 — CREATE API FILE

 app/api.py

```
from fastapi import FastAPI
```

```
from pydantic import BaseModel
```

```
from app.agent import Agent
```

```
from retrieval.search import HybridSearch
```

```
from tools.excel_agent import ExcelAgent
```

```
from ingestion.pdf_ingestor import load_pdf
```

```
from ingestion.excel_ingestor import load_excel
```

```
from ingestion.pdf_table_extractor import extract_pdf_tables
```

```
import pandas as pd
```

```
app = FastAPI()
```

```
# -----
```

```
# SYSTEM INIT (RUN ON START)
```

```
# -----
```

```
rag = HybridSearch()
```

```
#  Load PDF
```

```
docs = load_pdf("data/raw/sample.pdf")
```

```
rag.index(docs)
```

```
#  Excel Agent
```

```
excel_agent = None
```

```
try:
```

```
    df = load_excel("data/raw/sample.xlsx")
```

```
    excel_agent = ExcelAgent(df)
```

```
except:
```

```
    print("No Excel loaded")
```

```
#  PDF Table Agent
```

```
pdf_table_agent = None
```

```
try:
```

```
    tables = extract_pdf_tables("data/raw/sample.pdf")
```

```
    if tables:
```

```
        combined = pd.concat(tables, ignore_index=True)
```

```
        pdf_table_agent = ExcelAgent(combined)
```

except:

```
    print("No PDF tables found")
```

 Create Agent

```
agent = Agent(  
    rag_system=rag,  
    excel_tool=excel_agent,  
    pdf_table_tool=pdf_table_agent  
)
```

API SCHEMA

```
class Query(BaseModel):
```

```
    query: str
```

MAIN ENDPOINT

```
@app.post("/ask")
```

```
def ask(q: Query):
```

```
    answer = agent.run(q.query)
```

```
    return {
```

```
"query": q.query,  
"answer": answer  
}
```

✅ STEP 3 — RUN API

```
uvicorn app.api:app --reload
```

✅ STEP 4 — TEST (3 OPTIONS)

✅ Option 1 — Browser Swagger UI

```
http://127.0.0.1:8000/docs
```

✅ Option 2 — CURL

```
curl -X POST "http://127.0.0.1:8000/ask" \  
-H "Content-Type: application/json" \  
-d '{"query": "average sales"}
```

✅ Option 3 — Python test

```
import requests
```

```
res = requests.post(  
    "http://127.0.0.1:8000/ask",  
    json={"query": "what is ai"}  
)
```

```
print(res.json())
```

✅ FINAL FLOW

Request (POST /ask)



FastAPI



Agent System



Planner + Tools + Memory + Eval



JSON Response

✅ WHAT YOU BUILT

Backend AI system (API only)

✅ SYSTEM LEVEL NOW

- RAG ✅
 - Excel Agent ✅
 - PDF + OCR ✅
 - Multi-step Agent ✅
 - Memory ✅
 - Evaluation ✅
 - Logging ✅
 - ✅ FastAPI backend
-

✅ INDUSTRY EQUIVALENT

This is same architecture as:

Backend of ChatGPT / Copilot

✅ OPTIONAL (VERY IMPORTANT)

Later you can add:

- React UI
 - Postman testing
 - Docker deployment
-

✅ FINAL

- 👉 Clean + correct choice
 - 👉 Not bloated
 - 👉 Production-style
-

If next you want: 👉 Docker + deployment (real production)

👉 say: *day 15* 🚀